Reflective Journal – Module 9: Ensemble Methods

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1. Introduction

In this lab, I learned about ensemble methods and how combining multiple machine learning models can improve predictions. The focus was to understand the difference between a single Decision Tree and a Random Forest. I learned about the wisdom of the crowd, which means that using a group of models to make a prediction can sometimes be more reliable than depending on just one model. This lab aimed to train both models, compare their accuracy, and understand how feature importance helps identify the most useful information for making predictions.

2. Description of the Experience

During this lab, I used Python, Pandas, and Scikit-learn to work with the Iris flower dataset. I divided the data into training and testing sets, with 70% used for training and 30% used for testing. First, I trained a single Decision Tree classifier and evaluated its accuracy. The model achieved 100% accuracy on the test data.

Next, I created a Random Forest model using 100 decision trees. I trained the model and tested its predictions using the same data. The Random Forest also achieved 100% accuracy. I compared both models and saw that they performed equally well in this lab. I also created a feature importance plot to see which measurements were most useful for classifying the Iris flowers.

3. Personal Reflection and Learning

One of the most important things I learned from this lab was how ensemble methods work. Before this lab, I understood how a single Decision Tree makes predictions, but I had not fully understood how combining multiple trees could create a stronger model. I learned that a Random Forest uses many decision trees and combines their predictions to make a final decision.

I was surprised that both models achieved the same 100% accuracy. I originally expected the Random Forest to perform better because it uses 100 trees instead of one. I learned that adding more models doesn’t guarantee increase in accuracy because a simpler model can sometimes perform just as well, depending on the dataset.

Another important lesson was understanding feature importance. I learned that not every feature contributes equally to a model's predictions. Petal length and petal width were more important

than the other measurements when classifying the Iris flowers. This helped me understand how machine learning can identify patterns in data that might not be obvious at first.

4. Challenges, Improvements, and Questions

One challenge I faced was understanding how Random Forest combines the predictions of multiple decision trees. I also had to understand how to extract feature importance and display it using a bar chart. Working through the code step by step helped me understand how the model identifies important features.

5. Conclusion

Overall, Module 9 helped me improve my understanding of ensemble methods and how they are used in machine learning. My biggest takeaway was learning that combining multiple models can make predictions more reliable, but it does not guarantee better accuracy every time. Comparing the Decision Tree and Random Forest helped me see how two different approaches can produce the same results. I also learned how feature importance can help identify which parts of a dataset have the greatest influence on predictions.